Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)			any **two** of following observations for (1) each • both release misty or steamy fumes • only sodium bromide gives orange fumes / orange solution • only sodium bromide gives pungent smell (of sulfur dioxide)	2					2
				bromide is more easily oxidised than chloride / sulfuric acid is a strong enough oxidising agent to oxidise bromide but not strong enough to oxidise chloride		1		3		
	(b)	(i)		$K_a = \frac{[H^+][F^-]}{[HF]}$		1		1		
		(ii)		$K_a \times [HF] = [H^+]^2$ (1) $[H^+] = \sqrt{0.100 \times 7.20 \times 10^{-4}} = 0.00849$ (1) pH = − log $[H^+]$ = 2.07 (1)		3		3	3	

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		(iii)		vertical part of titration curve is between 6.5 (allow ± 0.5) and 11 (allow ± 1) / vertical part lies in alkaline region (1) select an indicator whose colour change is complete in the vertical region (1)	2			2		2
		(iv)	I	$HF \rightleftharpoons H^+ + F^-$ / reversible reaction (or equilibrium) (1) addition of a small amount of acid reacts with fluoride ions / shifts equilibrium to left (1) addition of a small amount of base removes hydrogen ions and these are replaced by HF dissociating / shifting equilibrium to the right / hydroxide ions react with HF molecules (1)	3			3		
			II	concentration of acid equal to concentration of salt (1) pH = − log [7.2×10^{-4}] = 3.14 (1)		2		2	1	
				Question 8 total	**7**	**7**	**0**	**14**	**4**	**4**